

Classwork 3

Quadratics and the discriminant

Grades: [A] [M] [E]

Answers · tutor's working, not an official mark scheme

Show all working. Give answers to 3 decimal places unless the question asks for an exact answer.

Part A · Solve, count, build

A1 Solve $(2x - 1)(x + 4) = 0$. [A]

$$x = \frac{1}{2} \quad \text{or} \quad x = -4$$

A2 Solve $x^2 - 3x = 28$. [A]

$$x^2 - 3x - 28 = 0, \quad (x - 7)(x + 4) = 0, \quad x = 7 \text{ or } x = -4$$

A3 Solve $3x^2 + 2x - 4 = 0$. [A]

$$x = \frac{-2 \pm \sqrt{(2)^2 - 4(3)(-4)}}{2(3)} = \frac{-2 \pm \sqrt{52}}{6}$$

$$x = 0.869 \text{ or } x = -1.535$$

A4 Find the discriminant of $x^2 - 4x + 7 = 0$ and say what it tells you. [A]

$$(-4)^2 - 4(1)(7) = -12 < 0, \text{ so there are no real roots}$$

A5 Write a quadratic equation with roots $\frac{2}{3}$ and -1 , with whole-number coefficients. [A]

$$(3x - 2)(x + 1) = 0, \quad 3x^2 + x - 2 = 0$$

Part B · Find the unknown

B1 $3x^2 + kx + 12 = 0$ has equal roots. Find k . [M]

$$k^2 - 4(3)(12) = 0, \quad k^2 = 144, \quad k = 12 \text{ or } k = -12$$

B2 $x^2 + 8x + p = 0$ has two different real roots. Find the values of p . [M]

$$64 - 4p > 0, \quad -4p > -64, \quad p < 16$$

B3 A parabola crosses the x -axis at -3 and 2 and passes through $(1, -8)$. Find its equation. [M]

$$y = a(x + 3)(x - 2); \quad \text{at } (1, -8): \quad -8 = a(4)(-1), \quad a = 2$$

$$y = 2(x + 3)(x - 2) = 2x^2 + 2x - 12$$

Part C · Real roots for every k

- C Show that $x^2 - (k + 3)x + 3k = 0$ has real roots for every value of k , and find when the roots are equal. **[E]**

Words you may use

discriminant · perfect square · never negative · therefore · equal roots

Sentence starters (optional): *The discriminant is ... This is a perfect square, so ... The roots are equal when*

...

$$\Delta = (-(k + 3))^2 - 4(1)(3k) = k^2 + 6k + 9 - 12k \quad \text{MP1}$$

$$= k^2 - 6k + 9 = (k - 3)^2 \quad \text{MP2}$$

A square is never negative, so $\Delta \geq 0$: the roots are real for every k . MP3

The roots are equal when $\Delta = 0$, that is when $k = 3$ (then $(x - 3)^2 = 0$).

MP1 is r; MP2 with the reasoning of MP3 is t.

Extension

Solve $x^4 - 5x^2 + 4 = 0$.

$$\text{Let } u = x^2: (u - 1)(u - 4) = 0, \quad x^2 = 1 \text{ or } 4, \quad x = \pm 1 \text{ or } x = \pm 2$$